FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Chemistry — Class IX (SSC-I) | Paper: English Medium

Syllabus: Ch 1 (Nature of Chemistry) | Ch 2 (Matter) | Ch 3 (Atomic Structure) | Ch 4 (Periodic Table) | Ch 5 (Chemical Bonding) | Ch 6 (Stoichiometry) | Ch 7 (Electrochemistry) | Ch 8 (Energetics) | Ch 9 (Chemical Equilibrium)

Total Marks: 65 **Time Allowed: 2h 30m** **Version: 3**

Atomic Masses (amu): H=1 · Li=7 · C=12 · N=14 · O=16 · Na=23 · Mg=24 · Al=27 · S=32 · Cl=35.5 · K=39 · Ca=40 · Mn=55 · Fe=56 · Cu=63.5 · Zn=65 · Br=80 · Ag=108 | $N_A = 6.022 \times 10^{23}$

General Instructions:

1. Write your roll number and section clearly on the answer sheet. Do not write your name.
2. Section A is compulsory. Encircle the correct option on the question paper itself.
3. In Section B, attempt **all 11** questions. For each question, attempt either the main question **OR** its alternative.
4. In Section C, attempt **all 4** questions. For each question, attempt either the main question **OR** its alternative.
5. Show all working in Sections B and C. Use black or blue ink only; pencil for diagrams only.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question	A	B	C	D	Answer
1	The branch of chemistry that studies the heat changes and the rates of chemical reactions is:	Physical chemistry	Organic chemistry	Analytical chemistry	Nuclear chemistry	ⒶⒷⒸⒹ
2	The direct change of a substance from the gaseous state to the solid state (without becoming a liquid) is called:	Sublimation	Deposition	Condensation	Freezing	ⒶⒷⒸⒹ
3	The maximum number of electrons that can be accommodated in a *d* sub-shell is:	2	6	10	14	ⒶⒷⒸⒹ
4	The electronic configuration of a chloride ion (Cl^-, Z = 17) is:	2, 8, 7	2, 8	2, 8, 8, 1	2, 8, 8	ⒶⒷⒸⒹ
5	Which of the following atoms has the *largest* atomic size?	H	Li	K	Na	ⒶⒷⒸⒹ
6	Which element is chemically inert because it has a completely filled outermost shell?	Sodium	Chlorine	Oxygen	Argon	ⒶⒷⒸⒹ
7	In which of the following substances is a *metallic* bond present?	Copper	NaCl	Diamond	HCl	ⒶⒷⒸⒹ
8	The mass of 0.25 mole of calcium carbonate ($CaCO_3$) is: (Ca = 40, C = 12, O = 16)	12.5 g	25 g	50 g	100 g	ⒶⒷⒸⒹ
9	The total number of *oxygen atoms* present in one mole of carbon dioxide (CO_2) is:	6.022×10^{23}	3.011×10^{23}	2	1.204×10^{24}	ⒶⒷⒸⒹ
10	The oxidation state of chlorine in the chlorate ion (ClO_3^-) is:	+5	+1	+3	+7	ⒶⒷⒸⒹ
11	The minimum amount of energy that colliding molecules must possess to start a chemical reaction is called:	Bond energy	Activation energy	Enthalpy change	Kinetic energy	ⒶⒷⒸⒹ
12	A reversible reaction can reach a state of equilibrium only if it is carried out in:	an open container	the presence of a catalyst in every case	a closed system	conditions of high pressure only	ⒶⒷⒸⒹ

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt all 11 questions. For each pair, attempt either the main question OR its alternative.

Part	Question	Marks	OR	OR Question	Marks
2(i)	Define **analytical chemistry** and **physical chemistry**. State one thing that each of them studies.	**3**	**OR**	What is **nuclear chemistry**? Give one beneficial use and one harmful effect of nuclear chemistry.	**3**
2(ii)	Differentiate between a **homogeneous mixture** and a **heterogeneous mixture**. Give one example of each.	**3**	**OR**	What is an **alloy**? Name one common alloy and state the main constituent metals present in it. Why is it classified as a solid solution?	**3**
	State the maximum number of electrons that			Define **valence shell** and **valence**	

2(iii)	State the maximum number of electrons that the **K, L and M shells** can hold, and write the rule (formula) that gives this maximum.	**3**	**OR**	Define **valence shell** and **valence electrons**. How many valence electrons does a sulphur atom (Z = 16) have?	**3**
2(iv)	Define **electron affinity**. State how it generally changes (a) across a period from left to right and (b) down a group.	**3**	**OR**	What is meant by **periodicity of properties**? Name any two properties that show a periodic variation in the periodic table.	**3**
2(v)	What is a **coordinate covalent bond**? Explain its formation in the hydronium ion (H_3O+), stating which species donates the electron pair.	**3**	**OR**	Using electron transfer, explain the formation of **magnesium oxide (MgO)** from a magnesium atom (Z = 12) and an oxygen atom (Z = 8).	**3**
2(vi)	Define **molar mass**. Calculate the molar mass of sodium carbonate (Na_2CO_3). (Na = 23, C = 12, O = 16)	**3**	**OR**	State Avogadro's number. Calculate the total number of **atoms** present in 0.5 mole of nitrogen gas (N_2).	**3**
2(vii)	Calculate the number of moles in 11 g of carbon dioxide (CO_2), and then the number of molecules present. (C = 12, O = 16)	**3**	**OR**	Balance the following equation and add state symbols: $Fe + H_2O$ (steam) → $Fe_3O_4 + H_2$	**3**
2(viii)	Determine the oxidation state of the central atom in: (a) Mn in MnO_2 (b) Cl in $HClO_4$ (c) P in H_3PO_4. Show working.	**3**	**OR**	Define a **redox reaction**. In $2Mg + O_2 \rightarrow 2MgO$, identify the species oxidized and the species reduced, giving the change in oxidation state of each.	**3**
2(ix)	What is **corrosion**? Explain, with the principle involved, how (a) painting and (b) oiling or greasing protect iron from corrosion.	**3**	**OR**	Why does aluminium resist corrosion even though it is a fairly reactive metal? Explain the protective role of its oxide layer.	**3**
2(x)	A spark or flame is needed to start the combustion of methane even though the reaction is exothermic overall. Using the idea of **activation energy**, explain why energy must be supplied to begin such a reaction.	**3**	**OR**	Define a **catalyst**. Name one industrial catalyst and the reaction in which it is used, and state how a catalyst affects (a) the rate of reaction and (b) the overall enthalpy change (ΔH).	**3**
2(xi)	Differentiate between **physical equilibrium** and **chemical equilibrium**. Give one example of each.	**3**	**OR**	For a reversible reaction in a sealed flask, describe how the **rate of the forward reaction** and the **rate of the reverse reaction** change from the moment of mixing until equilibrium is established.	**3**

SECTION C — Long Questions (4 × 5 = 20 Marks) — Attempt all 4

Part	Question	Marks	OR	OR Question	Marks
3(i)	(a) Define electronic configuration. Write the electronic configuration (in shells) of the following atoms and, for each, state its group and period: (i) carbon (Z = 6) (ii) sulphur (Z = 16). [3] (b) An atom X has 12 protons and 12 electrons. State (i) whether it is a metal or a non-metal and why, (ii) the charge on the ion it forms, and (iii) its valency. [2]	**5**	**OR**	(a) Define ionization energy. Explain why the first ionization energy of sodium is much lower than that of chlorine. [3] (b) Arrange the elements Na, Mg and Al in order of *increasing* atomic size and justify your order. [2]	**5**
3(ii)	(a) Define empirical formula and molecular formula. A compound has the empirical formula CH_2O and a molecular mass of 180 amu. Determine its molecular formula. (C = 12, H = 1, O = 16) [3] (b) Calculate the number of molecules present in 3.6 g of water (H_2O). (H = 1, O = 16) [2]	**5**	**OR**	(a) Define the mole. Calculate: (i) the number of moles in 8 g of methane (CH_4) and (ii) the mass of 2 moles of nitrogen gas (N_2). (C = 12, H = 1, N = 14) [3] (b) Balance the equation $Al + Fe_2O_3 \rightarrow Al_2O_3 + Fe$ (thermite reaction) and state why it is classified as a redox reaction. [2]	**5**
3(iii)	(a) Define oxidation and reduction in terms of electron transfer. In $Zn + 2AgNO_3 \rightarrow Zn(NO_3)_2 + 2Ag$, identify the oxidizing agent and the reducing agent, giving the change in oxidation number of each. [3] (b) Using the rules for oxidation states, determine the oxidation state of (i) Mn in the manganate ion MnO_4^{2-} and (ii) S in the sulphite ion SO_3^{2-}. Show working. [2]	**5**	**OR**	(a) Define an oxidizing agent and a reducing agent. For $MnO_2 + 4HCl \rightarrow MnCl_2 + Cl_2 + 2H_2O$, identify the oxidizing agent and the reducing agent, justifying your answer with oxidation-state changes. [3] (b) Balance the equation $Fe + Cl_2 \rightarrow FeCl_3$ and state which element is oxidized. [2]	**5**
	(a) Using the bond energies below, calculate ΔH for the reaction and state whether it is exothermic or endothermic:			(a) Using the bond energies below, calculate ΔH for the reaction and state whether it is exothermic or endothermic:	

3(iv)	$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$. Bond energies (kJ/mol): C–H = 414, O=O = 498, C=O = 803, O–H = 463. [3] (b) The reaction $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ is carried out in a sealed vessel. (i) What does the symbol $\rightleftharpoons$ tell you about the reaction? (ii) Does the reaction stop when equilibrium is reached? Explain. [2]	**5**	**OR**	$H_2(g) + Br_2(g) \rightarrow 2HBr(g)$. Bond energies (kJ/mol): H–H = 436, Br–Br = 193, H–Br = 366. [3] (b) State whether each of the following is exothermic or endothermic and give the sign of ΔH: (i) respiration in the body (ii) photosynthesis in green plants. [2]	**5**

Invigilator's Signature: ____________________ Candidate's Roll No.: ____________________

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— End of Question Paper —